Fiscal Year:	FY 2020	**Task Last Updated:**	FY 07/14/2021
PI Name:	Warren, Warren Ph.D.		
Project Title:	Compaction and Sound in Granular Matter		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Durham	**State:**	NC
Zip Code:	27708	**Congressional District:**	1
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	07/11/2018	**End Date:**	01/12/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AD38G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (December 2018): Original Principal Investigator Dr. Robert Behringer passed away in July 2018; project continues with Dr. Warren Warren as PI. Note also that original grant was NNX10AU01G, 10/1/2010-1/12/2015, entitled "Compaction and Sound Transmission in Dense Granular Media"; continuation grant established as "Compaction and Sound in Granular Matter," NNX15AD38G, 1/13/2015-1/12/2020. This project consists of a collaboration between researchers in France, Germany, and the United States, involving several leading teams in the field of granular materials/soft matter coming from these countries. The research program will exploit reduced-gravity conditions in order to understand the states of granular matter near jamming. Microgravity conditions are essential for achieving uniform granular samples, particularly near the jamming transition. Key scientific goals include: 1) Understanding the relation between classical ideas, including random-close and random-loose densities, and recent advances such as shear jamming, isotropic jamming, etc.; 2) A determination of critical properties of packings near jamming; 3) An investigation of sound transmission for packings near jamming. These studies will exploit both 2D and 3D techniques. The 3D techniques will exploit both photoelasticity and laser scanning devices. Two-dimensional photoelastic techniques have been extensively developed by the US Principal Investigator (PI), Robert Behringer, and his contribution will involve both quasi-static and acoustic studies of granular materials made from quasi-2D photoelastic particles. In addition, the US part of the collaboration will develop and use 3D systems based on laser-scanning techniques. Several technical goals for this project include: 1) further development of 3D scanning and reconstruction techniques; 2) further development of facilities for studying granular systems onboard the International Space Station (ISS), on sounding rockets, and on parabolic flights; 3) extension of highly quantitative photoelastic techniques from 2D to 3D; 4) the combining of photoelastic approaches and conventional acoustic measurement techniques; 5) the development of super-low friction substrates for acoustic measurements in 2D; 6) continued investigations that seek to understand jamming in 2D systems, where gravitational effect can be mitigated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It has been known for a long time that stresses in granular materials stored in a container do not develop in the same way as the hydrostatic stresses in a fluid stored in the same container. Understanding the stresses in static, and slowly-moving, and dynamic dense granular materials is a research area of considerable importance, impacting such industries as pharmaceutical, cement, mining, chemical, and silo design/construction.
Task Progress:	This grant was awarded to Professor Robert Behringer, who passed away suddenly in July 2018. Members of his research group continued work for a limited time, supported by grant funds. Student support, which of course is restricted to US citizens and permanent residents, was not expended after 2018 as the students moved on to different research projects. With NASA support, the project investigated two experiments exploring stability and failure in granular media. First, an experiment was built to explore the passage of a single grain that is driven through quasi-two-dimensional packing disks; single-grain intruder dynamics elucidate the most fundamental mechanisms of stability formation and failure [1-4]. It was found that basal friction between the disks and the glass substrate on which they sit strongly controls the dynamics of the grains; with basal friction, stick-slip dynamics are observed, and without basal friction, clogging dynamics are observed. In stick-slip dynamics, the intruder grain experiences sticking periods in which the granular medium is stable [5]; sticking periods are separated by relatively short duration intermittent slip events in which the granular medium yields under a critical force that varies from sticking period to sticking period. In clogging dynamics, the intruder freely flows through the granular medium but only occasionally gets stopped when the medium randomly forms a stable arch reaching to the system boundaries [6]. Thus, the system exhibits physical dynamics akin to clogging of granular media flowing through restricted apertures if the basal support is not present to stabilize the medium under increasing loads. This work was presented at the Granular Matter Gordon Research Conference in 2018 and the APS March Meeting 2019. Second, a three-dimensional packing of density-matched hydrogel spheres or plastic balls (frictionless and frictional limits, respectively) suspended in fluid was placed under boundary-driven cyclic shear, since many simulations have observed non-trivial behaviors for three-dimensional granular media under cyclic shear, such as limit cycles of particle positions in the steady-state from cycle to cycle [7-9]. For frictional plastic balls, the maximum pressure at the amplitude of shear was found to vary dramatically from cycle to cycle, similarly to the chaotic behavior observed in [10]. With low-friction hydrogels, the maximum pressure at the amplitude of shear was repeatable under the same straining conditions as the plastic balls for high packing densities. However, the hydrogels exhibited a different steady state behavior at particle densities close to the densities in which jamming never occurs: the system develops an initial maximum pressure that decreases with cycles to a steady state value, presumably through annealing by the action of the cyclic shear. This work was presented at the APS March Meeting 2019. References [1] J. Geng, G. Reydellet, E. Clément, and R. P. Behringer, Phys. D Nonlinear Phenom. 182, 274 (2003). [2] A. Tordesillas, J. E. Hilton, and S. T. Tobin, Phys. Rev. E 89, 042207 (2014). [3] R. Bharadwaj, C. Wassgren, and R. Zenit, Phys. Fluids 18, 043301 (2006). [4] A. Seguin, C. Coulais, F. Martinez, Y. Bertho, and P. Gondret, Phys. Rev. E 93, 012904 (2016). [5] S. Nasuno, A. Kudrolli, and J. P. Gollub, Phys. Rev. Lett. 79, 949 (1997). [6] C. C. Thomas and D. J. Durian, Phys. Rev. Lett. 114, 178001 (2015). [7] J. R. Royer and P. M. Chaikin, Proc. Natl. Acad. Sci. 112, 49 (2014). [8] S. Farhadi, A. Z. Zhu, and R. P. Behringer, Phys. Rev. Lett. (2015). [9] J. Ren, J. A. Dijksman, and R. P. Behringer, Phys. Rev. Lett. 110, 018302 (2013). [10] I. Regev, T. Lookman, and C. Reichhardt, Phys. Rev. E 88, 062401 (2013).

Bibliography Type:	Description: (Last Updated: 06/13/2025)
Abstracts for Journals and Proceedings	Wang M, Wang D, Socolar J, Zheng H, Behringer RP. "Jamming by shear in a dilating granular system." APS March Meeting 2019, Boston, MA, March 4-8, 2019. Session E56: In the memory of Bob Behringer: Statistical and Nonlinear Physics. Bulletin of the American Physical Society. 2019 Mar;64(2):Abstract: E56.00002. https://meetings.aps.org/Meeting/MAR19/Session/E56.2 , Mar-2019
Abstracts for Journals and Proceedings	Wang D, Zheng H, Bares J, Behringer RP. "System Size Dependence in Shear Jamming of 2D Frictional Grain." APS March Meeting 2019, Boston, MA, March 4-8, 2019. Session E56: In the memory of Bob Behringer: Statistical and Nonlinear Physics. Bulletin of the American Physical Society. 2019 Mar;64(2):Abstract: E56.00003. https://meetings.aps.org/Meeting/MAR19/Session/E56.3 , Mar-2019
Abstracts for Journals and Proceedings	Chen DZ, Kozlowski R, Behringer RP. "Abstract: E56.00005 : Experimental observation of shear jamming in 3D granular materials." APS March Meeting 2019, Boston, MA, March 4-8, 2019. Session E56: In the memory of Bob Behringer: Statistical and Nonlinear Physics. Bulletin of the American Physical Society. 2019 Mar;64(2):Abstract: E56.00005. https://meetings.aps.org/Meeting/MAR19/Session/E56.5 , Mar-2019
Abstracts for Journals and Proceedings	Zhao Y, Bares J, Zheng H, Zadeh AA, Socolar J, Behringer RP. "Nontrivial plasticity of a shear-jammed granular system." APS March Meeting 2019, Boston, MA, March 4-8, 2019. Session E56: In the memory of Bob Behringer: Statistical and Nonlinear Physics. Bulletin of the American Physical Society. 2019 Mar;64(2):Abstract: E56.00006. https://meetings.aps.org/Meeting/MAR19/Session/E56.6 , Mar-2019
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Abstracts for Journals and Proceedings	Kozlowski R, Chen DZ, Zheng H, Socolar J, Kondic L, Daniels K, Behringer RP. "Dynamics of a grain-scale intruder in a granular system with and without basal friction." APS March Meeting 2019, Boston, MA, March 4-8, 2019. Session E56: In the memory of Bob Behringer: Statistical and Nonlinear Physics. Bulletin of the American Physical Society. 2019 Mar;64(2):Abstract: E56.00011. https://meetings.aps.org/Meeting/MAR19/Session/E56.11 , Mar-2019
Abstracts for Journals and Proceedings	Zhao Y, Bares J, Behringer RP. "Interlocking of Sheared Non-convex Hexapods." APS March Meeting 2019, Boston, MA, March 4-8, 2019. Bulletin of the American Physical Society. 2019 Mar;64(2):Abstract: X48.00001. https://meetings.aps.org/Meeting/MAR19/Session/X48.1 , Mar-2019
Articles in Peer-reviewed Journals	Zhao Y, Barés J, Zheng H, Socolar JE, Behringer RP. "Shear-jammed, fragile, and steady states in homogeneously strained granular materials." Physical review letters. 2019 Oct 11;123(15):158001. https://doi.org/10.1103/PhysRevLett.123.158001 , Oct-2019